Decompose a Goal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach reusable functions through distinct calls.

Provide full definitions, model positional binding and separate the printing task from the return-value preview.

Keep one habit: Look for executed bodies, correct parameters and a limited input contract.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Distinguish a **function definition** from a **function call**

Teach **definition**, **call**, **parameter** and positional **argument** as distinct ideas. Keep **input**s scalar and avoid mutable-list aliasing in this first **function** lesson.

Reference: https://docs.python.org/3/tutorial/controlflow.html#defining-**function**s. These **function**s print results; reusable return values and local state are taught in Week 9.

By the end, I can **decompose** a calculation into a **function** with explicit **argument**s and a returned result.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/2, python-bridge/8. Reuse a familiar case for re**call** or modelling; verify that the case used to assess transfer has not already received feedback.
- Python 3 editor and **function**s.py
- **Parameter** cards labelled capacity and used
- **Call** cards and a table for expected and observed **output**

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

8. Subtraction is ordered: reversing these values would produce -8.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Precision in Algorithms

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

Concept to explain

Complex tasks become manageable when divided into testable **subproblem**s.

Term	Meaning
decompose	Break a goal into smaller parts with clear responsibilities and connections.
subproblem	A smaller part of a larger problem with its own inputs and intended result.
dependency	Something a step needs from another step before it can proceed correctly.
interface	An agreed boundary specifying how two parts exchange information or actions.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: re**call** 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach definitions, positional calls and explicit interfaces

> Keep the printing cupboard task distinct from the optional returned-label preview.

State the cupboard job and input assumptions

A fictional supply cupboard records capacity and the number of occupied spaces. The remaining-space calculation is capacity minus used.

Instead of copying the same instructions for every cupboard, define one **function** and **call** it with different values. A **function** makes a procedure reusable; its name alone does not guarantee that its instructions are correct.

Our **input** contract requires nonnegative integer counts with used no greater than capacity. The example assumes these conditions; it does not validate them. The main cupboard **function** prints its result. The labelled extension shows a different interface: returning a value for the caller to use or display.

Read the complete correct script

```
def show_remaining(capacity, used):
    remaining = capacity - used
    print(remaining)
```

```
show_remaining(12, 4)
show_remaining(10, 7)
```

Explain definition, body and calls

def introduces the **definition**. The **function** name is followed by **parameter** names in parentheses and a colon. Four-space indentation marks the body. Executing the **definition** creates the **function** but does not run its body.

The two unindented **call**s then run it, producing 8 and 3 on separate lines. A blank line helps readability; indentation determines the block.

Trace positional arguments and output order

Execution event	What happens
Execute the definition	Create the function; no output yet
Call with 12 and 4	Match parameters; subtract; print 8
Call with 10 and 7	Match new values; subtract; print 3
After the second call	Continue with the next statement

In show_remaining(12, 4), 12 and 4 are **argument**s; capacity and used are **parameter**s.

Positional **argument**s match from left to right. show_remaining(4, 12) produces -8, showing why order matters; it also violates our written **input** contract. Use a fresh file with the **definition** before its **call**s.

> **Predict the output of show_remaining(9, 9) followed by show_remaining(9, 0). Name the parameter receiving each argument. What would be printed if both calls were removed?**

AI ACADEMY / AI-2/2 · 03 · INSPECT THE LESSON MODEL

Arguments matched in order

Call	Parameter values	Printed output
show_remaining(12, 4)	capacity 12; used 4	8
show_remaining(10, 7)	capacity 10; used 7	3

Each call runs the same body with the values supplied for that call.

Follow the labelled picture with the matching explanation. Open the vector diagram

Preview the separate returning function

A fictional library computes total spine labels needed for several equal shelves.

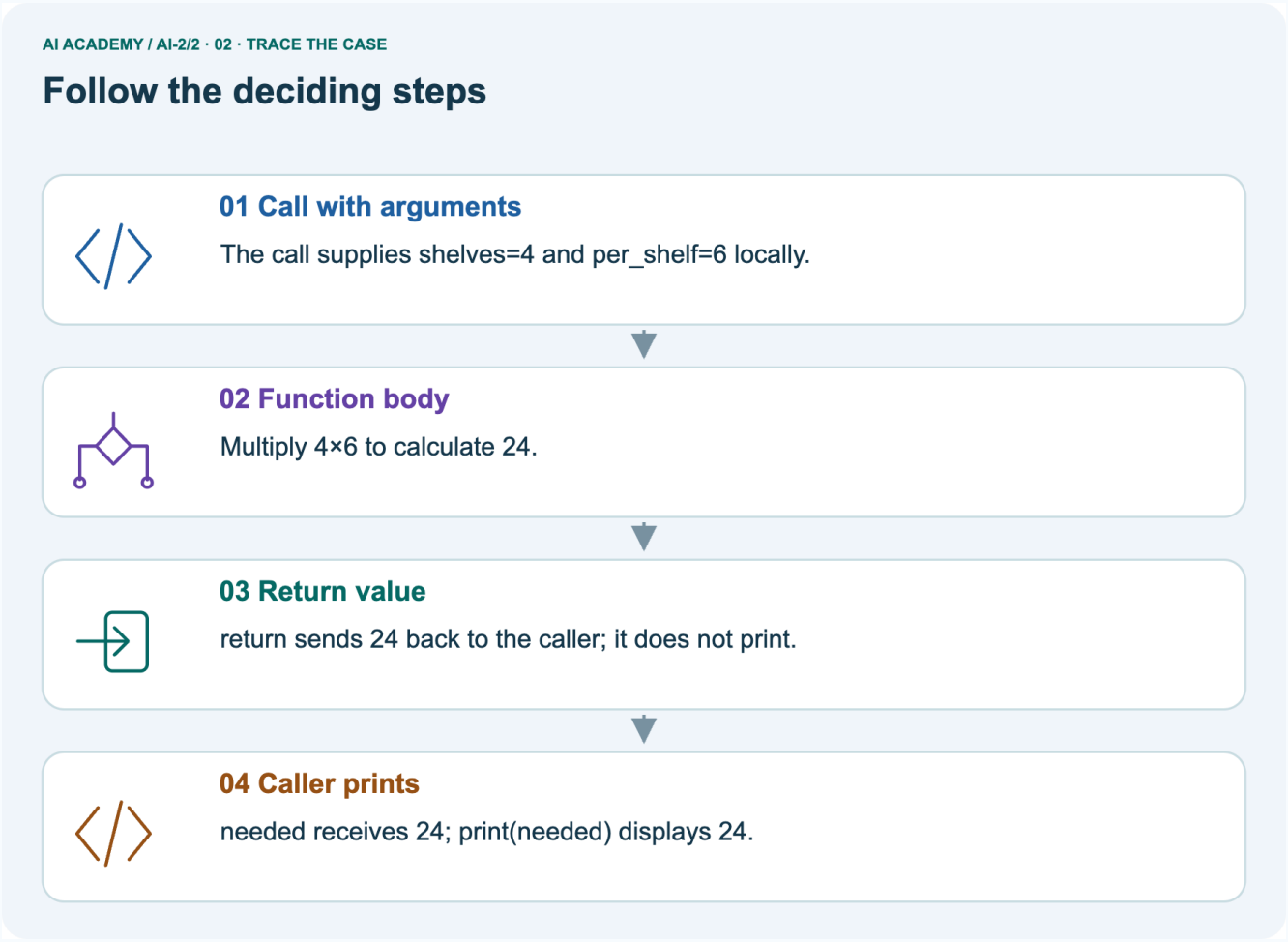
Define labels(shelves,per_shelf): return shelves*per_shelf.

Call needed=labels(4,6), then print(needed). No other side effects are defined.

```
def labels(shelves, per_shelf):
    return shelves * per_shelf

needed = labels(4, 6)
print(needed)
```

Trace return versus caller print



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Call with arguments	The call supplies shelves=4 and per_shelf=6 locally.
2. Function body	Multiply 4×6 to calculate 24.
3. Return value	return sends 24 back to the caller; it does not print.
4. Caller prints	needed receives 24; print(needed) displays 24.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Keep the input-type assumption visible

This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

Change labels per shelf to zero

Change: **Call** labels(4,0) instead.

Prediction: The returned result is 0, and the **call**er prints 0.

Reason: Four shelves times zero labels per shelf is zero.

Use pictures to check reasoning

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Goal: help a learner leave home ready.

2. Parts: check time; check weather; check bag; choose reminder.

3. **Dependency**: weather must be known before the clothing suggestion.

4. Test each part separately, then test the whole flow.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Exact explanation to model

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

Explain the added worked example

Key terms

Term	Meaning
Function	A named block of reusable instructions that runs when called
Definition	Code that creates a function and specifies its body
Call	An instruction that runs a function with supplied arguments
Parameter	A name in a function definition that receives an argument
Argument	A value supplied when calling a function

Model the reasoning aloud

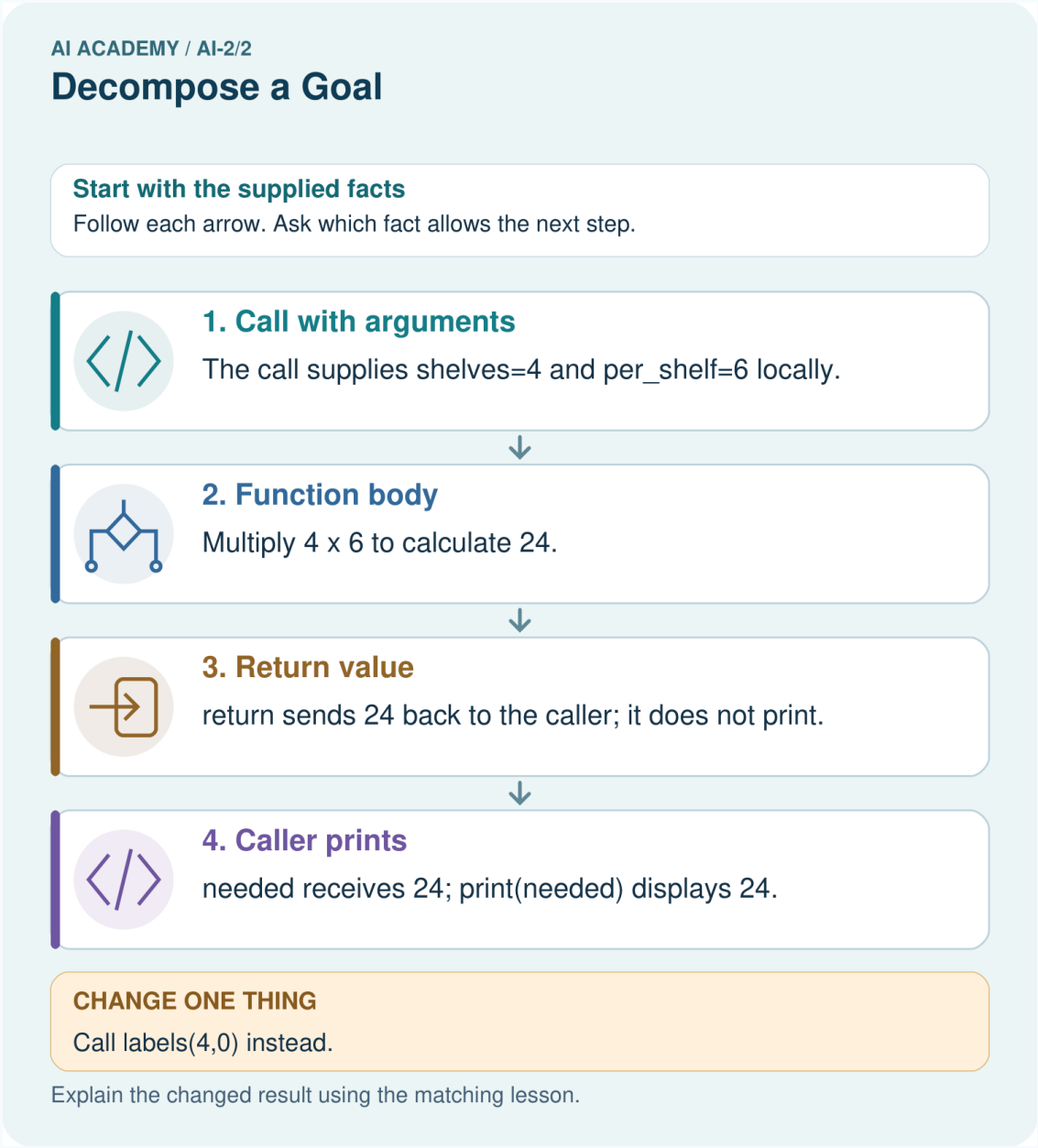
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Decompose a Goal

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

Expected reasoning and fading help

Teacher check: The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The **definition** creates the **function**; a **call** runs its body.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

What happens visibly if the **call**er stores needed but omits print(needed)?

The calculation returns 24 and needed stores it, but the shown program displays nothing because no print runs.

Changed-case reasoning: The returned result is 0, and the **call**er prints 0. Four shelves times zero labels per shelf is zero.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A class wants to recommend one of five library books.

Write the question to explain: **Decompose** the goal into checkable steps.

Reveal after **prediction**s: Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

Use the earlier changed question as optional later practice. Reserve Create a new function independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

The **call**s print 0 then 9. capacity receives 9 in both **call**s; used receives 9 then 0. Removing both **call**s leaves a **definition** with no printed **output**. Exit: show_remaining(10, 7) should print 3 but the faulty version prints 8.

Passing two cases does not establish correctness for all allowed **input**s or validate the **input** contract.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve ordered subtraction with the prerequisite.
7 to 20 minutes	Read the contract and label the definition and calls for W1.
20 to 33 minutes	Match argument cards to parameters and trace the worked calls.
33 to 45 minutes	Complete W2 and the guided question before running the file.
Session 2 start to 8 minutes	Retrieve why a definition alone prints nothing.
8 to 20 minutes	Run the fixed-number bug, complete W3 and rerun the earlier case after repair.
20 to 35 minutes	Collect the independent W4 function and predictions before hints.
35 to 45 minutes	Discuss W5 and use the exit to assess the limits of passing tests.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Correct show_remaining prints 8,3,0,9 for the four lab calls. The fixed 12-4 body prints 8 each time; preserve this observed fault and rerun after replacing it with capacity-used. A missing positional argument raises TypeError; swapped (4,12) gives -8 but violates the stated cupboard contract. The optional labels preview returns 24 for (4,6), 0 for (4,0), and prints only when its caller asks. Keep predictions separate from actual runs. Reserve Create a new function independently for an uncoached show_stock attempt.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Cut a familiar task into 4–6 cards. Order dependencies, name each **input** and **output**, and test whether replacing one card breaks the whole system.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Writing def runs the body immediately: The **definition** creates the **function**; a **call** runs its body.

Arguments and **parameter**s are interchangeable labels: **Parameter**s are names in the **definition**; **argument**s supply values at a **call**.

Swapping **argument**s never matters: Different **parameter** roles can give different results when values are swapped.

A correct first example proves the **function** is reusable: Fixed numbers can match one case while ignoring later **argument**s.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A **definition** matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Create a new function independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

The body prints boxes * 6 + loose. **Output**s are 20, 2 and 18. show_stock(2, 3) gives 15 because boxes and loose have different roles. Define once and use three **call**s; do not hard-code one answer.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automati**call**y an equivalent test.

W1 Label the parts

In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

Name show_remaining; **parameter**s capacity and used. **Argument**s 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

W2 Follow the execution order

Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

W3 Repair ignored arguments

The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

Faulty **output** 8; expected 9. Use capacity - used. Rerun show_remaining(12, 4), expecting 8, as well as the revealing **call**.

W5 State the input assumption

The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

No. used exceeds capacity and violates the contract. The **function** calculates with supplied values but does not check the contract. A separate validation check or guarded **function** is needed to reject invalid records.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

Additional transfer practice question

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

Reasoned practice answer

Collect a topic; use that topic to select matching records; pass a chosen record to display its title. Each step has a specific **input** and **output**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Decomposition is only making a longer task list.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Read request; validate required fields; filter by genre and pages<120; score remaining books; apply a stated tie/no-match policy; explain the result. Pass request, filtered IDs, scores and selected ID between stages.

Scoring first is not necessarily wrong if filtering still occurs, but it makes the constraint easier to overlook.

Explanations for the additional workbook examples

W1 Label the parts

W2 Follow the execution order

W3 Repair ignored arguments

W4 Create a new function independently

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

The body prints boxes * 6 + loose. **Output**s are 20, 2 and 18. show_stock(2, 3) gives 15 because boxes and loose have different roles. Define once and use three **call**s; do not hard-code one answer.

W5 State the input assumption

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Separate reader profile, book data, score calculation, ranking, and explanation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use two labelled **parameter** spaces and place **argument** cards left to right. Permit an oral execution trace alongside code.

Extension: add an if check inside show_remaining that prints Invalid when used exceeds capacity, otherwise prints the difference; test (4, 5) and (4, 4). This checks only that one constraint.

W4 must define one **function**, use both **parameter**s, **call** it three times and predict 20, 2, 18; the swapped case must give 15.

After support, retry show_tiles(packs, singles) with 4 tiles per pack: **call**s (2, 3), (0, 3), (2, 0) print 11, 3, 8; swapped (3, 2) prints 14. Record assistance and collect new **prediction**s before execution.

Plan the delayed check

Suggested return points in this level: ai-2/3, ai-2/5, ai-2/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 3: Branching Logic. Retrieve today's **evidence** and identify what carries forward and what is new.

The labelled extension distinguishes returned values from printed output; later project stages build on explicit function interfaces and fresh state for each **call**.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Distinguish a **function definition** from a **function call**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order.

Explain what happens if the file contains only the **definition**.

Reasoned practice answer: The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

Check every learner before discussion: Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The **definition** creates the **function**; a **call** runs its body.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: **Call** labels(4,0) instead. Require a **prediction** and a reason before comparing. Expected result: The returned result is 0, and the **call**er prints 0. Reason: Four shelves times zero labels per shelf is zero.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

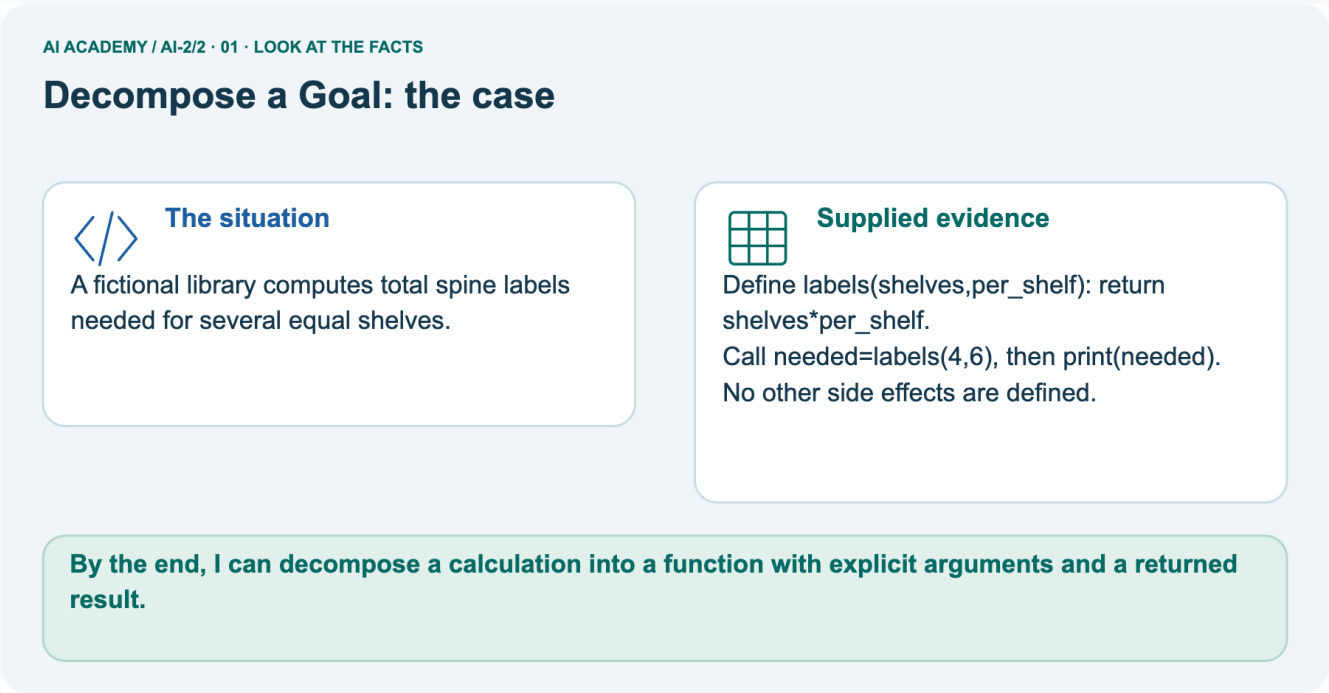
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Decompose a Goal: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

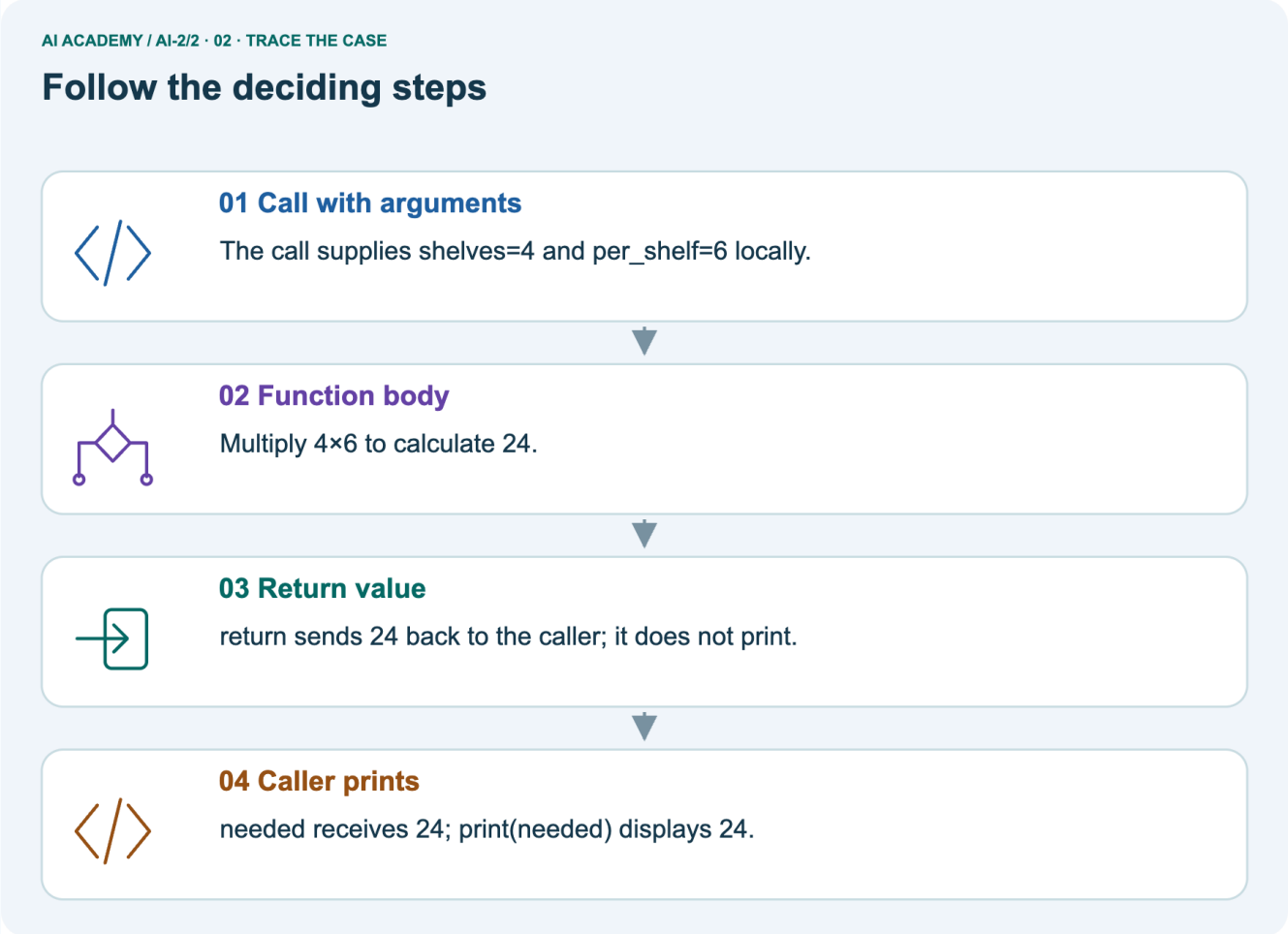
A fictional library computes total spine labels needed for several equal shelves. Supplied facts: Define labels(shelves,per_shelf): return shelves*per_shelf. **Call** needed=labels(4,6), then print(needed).

No other side effects are defined.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



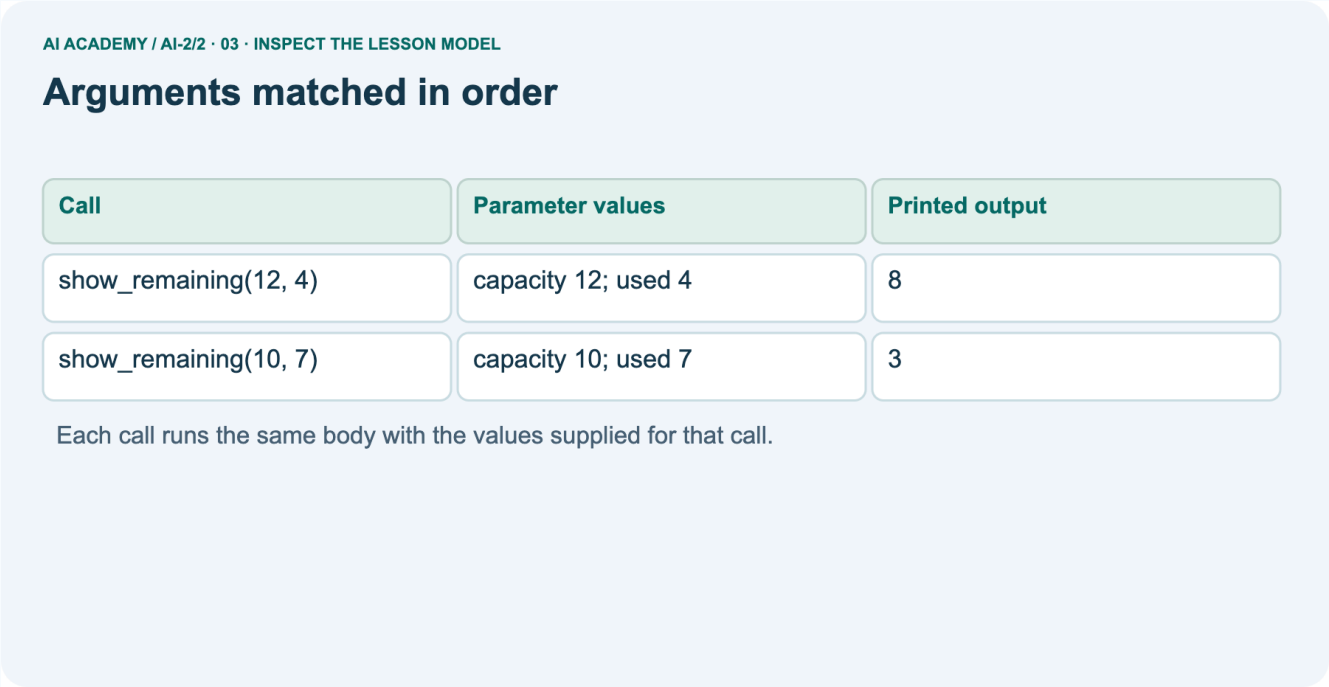
Follow the labelled picture with the matching explanation. Open the vector diagram

Call with **argument**s: The **call** supplies shelves=4 and per_shelf=6 lo**call**y. → **Function** body: Multiply 4×6 to calculate 24. → Return value: return sends 24 back to the **call**er; it does not print. → **Call**er prints: needed receives 24; print(needed) displays 24.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Arguments matched in order



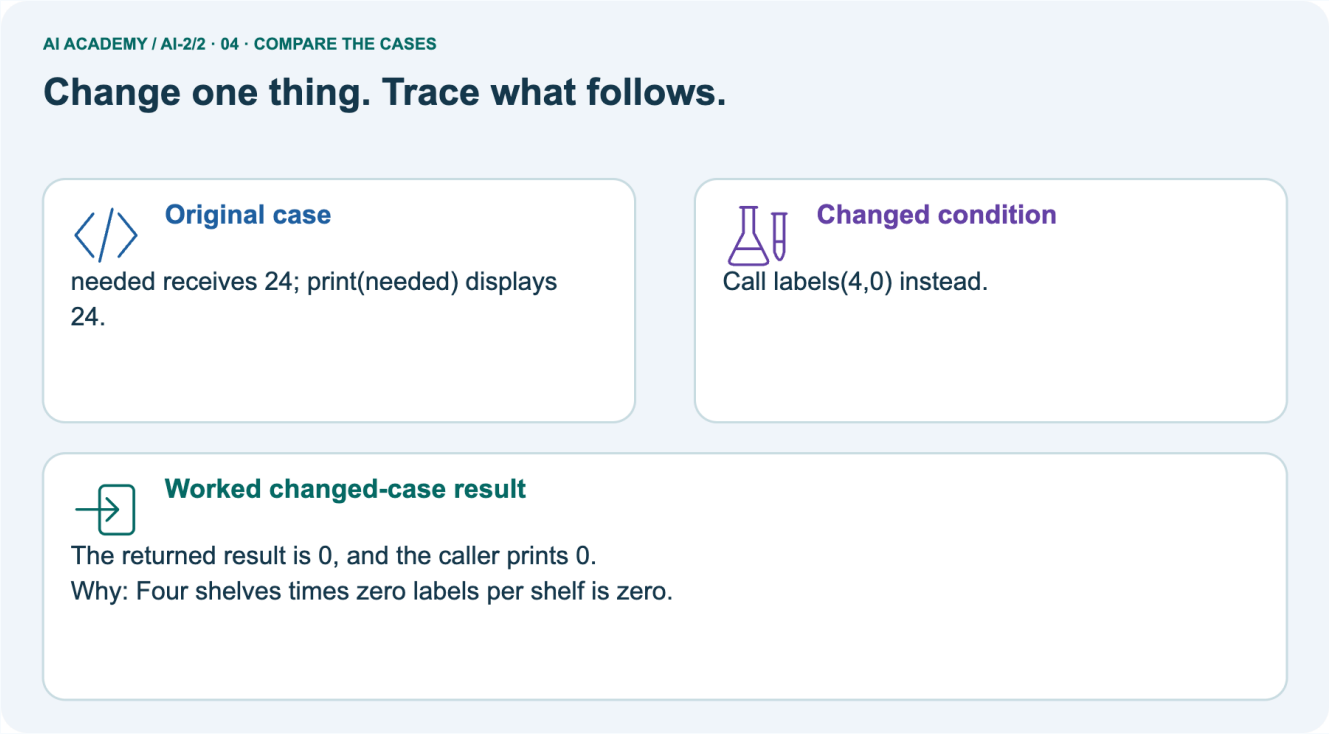
Follow the labelled picture with the matching explanation. Open the vector diagram

Arguments matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: needed receives 24; print(needed) displays 24. Changed condition: **Call** labels(4,0) instead. Expected result: The returned result is 0, and the **call**er prints 0.

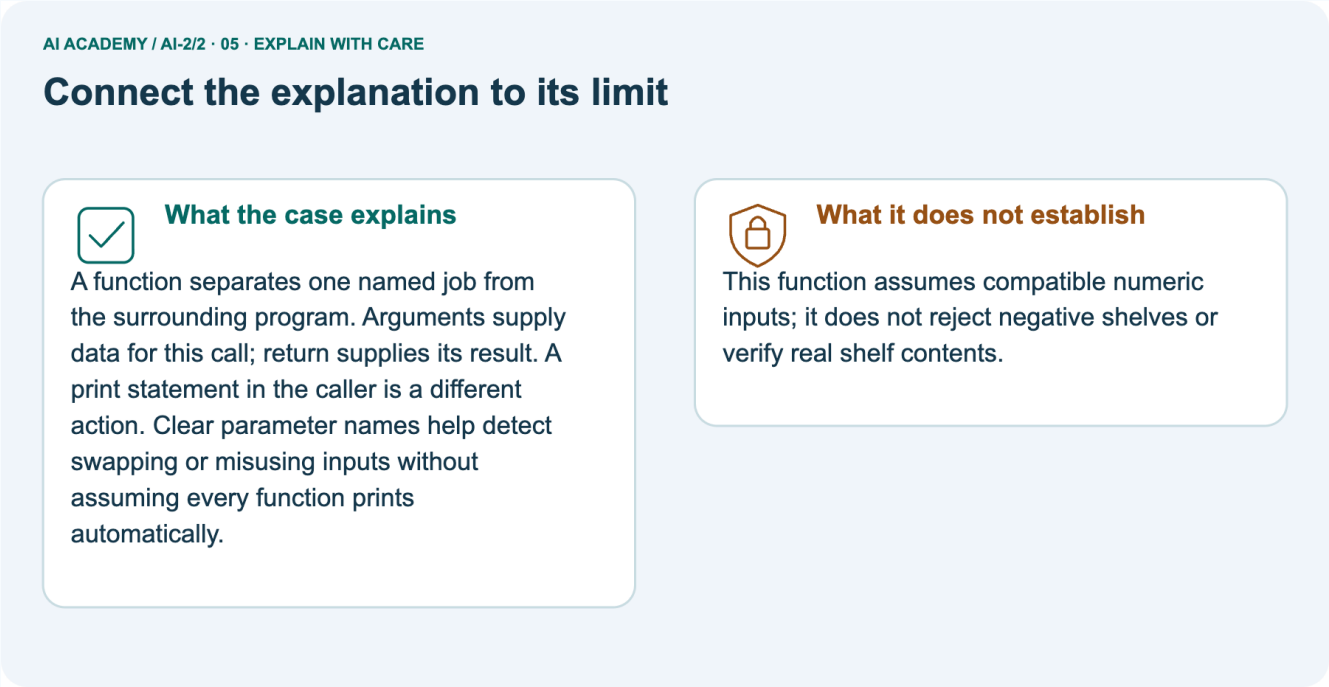
Reason: Four shelves times zero labels per shelf is zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Four shelves times zero labels per shelf is zero.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

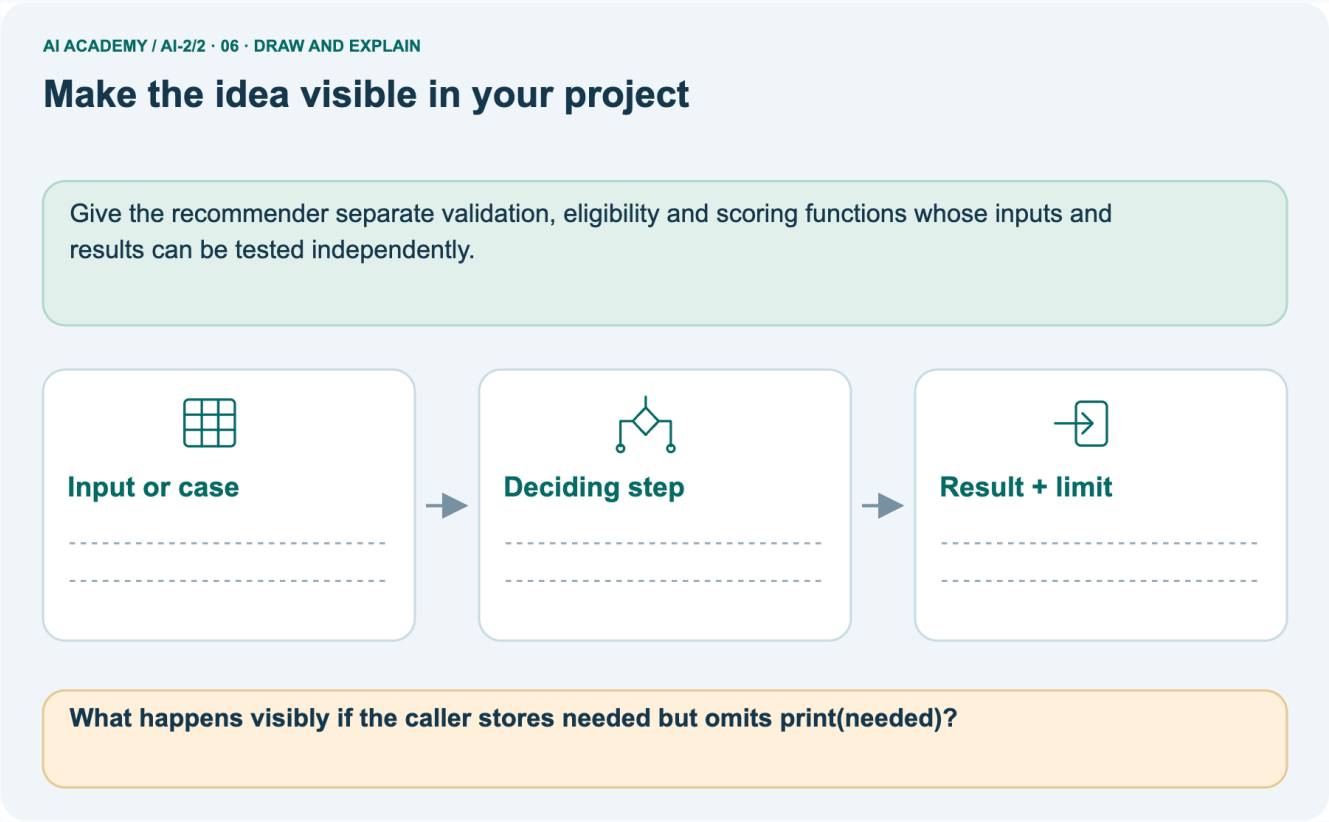
Limit: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently. Draw your **input**, deciding step, result and limit.

Supported question: What happens visibly if the **call**er stores needed but omits print(needed)?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The calculation returns 24 and needed stores it, but the shown program displays nothing because no print runs.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Decompose a Goal: the case

02. Follow the deciding steps

03. Arguments matched in order

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.